

MS101 – Makerspace
2023-24/I (Autumn Semester)
EE – Quiz 2

Answers to Numerical Questions

Ques.	Answers	Comments																																								
1	a) $R_C = 300\ \Omega$, $V_x = 9\text{ V}$ b) $R_B = 5200\ \Omega$																																									
2	a) <table border="1"><thead><tr><th>MN_A</th><th>MN_B</th><th>MP_C</th><th>LED</th><th>MN_A</th><th>MN_B</th><th>MP_C</th><th>LED</th></tr></thead><tbody><tr><td>OFF</td><td>OFF</td><td>OFF</td><td>0</td><td>OFF</td><td>OFF</td><td>ON</td><td>0</td></tr><tr><td>OFF</td><td>ON</td><td>OFF</td><td>0</td><td>OFF</td><td>ON</td><td>ON</td><td>1</td></tr><tr><td>ON</td><td>OFF</td><td>OFF</td><td>0</td><td>ON</td><td>OFF</td><td>ON</td><td>1</td></tr><tr><td>ON</td><td>ON</td><td>OFF</td><td>0</td><td>ON</td><td>ON</td><td>ON</td><td>1</td></tr></tbody></table> b) $(A + B) C'$	MN_A	MN_B	MP_C	LED	MN_A	MN_B	MP_C	LED	OFF	OFF	OFF	0	OFF	OFF	ON	0	OFF	ON	OFF	0	OFF	ON	ON	1	ON	OFF	OFF	0	ON	OFF	ON	1	ON	ON	OFF	0	ON	ON	ON	1	
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3	a) 19.67 mA (20 mA) b) 19.67 mA (20 mA)																																									
4	C																																									
5	a) $y = \text{map}(x, 240, 0, 100)$ b) i) New min val. = 396, New max val. = 1000 c) analogReference(INTERNAL)																																									
6	a) Period = 37.5 ms b) RPM = 122.95 c) $R = 78.64\text{ k}\Omega$ or $6.36\text{ k}\Omega$																																									